Q1

Student ID → xx ... 4 3 2 1

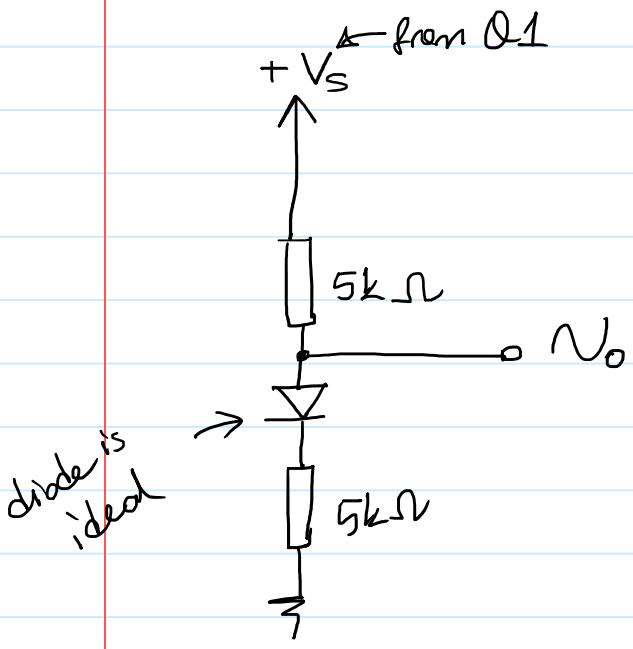
(last four digits)

$V_s = (3 + 1) + 1 = 5$

(fixed)

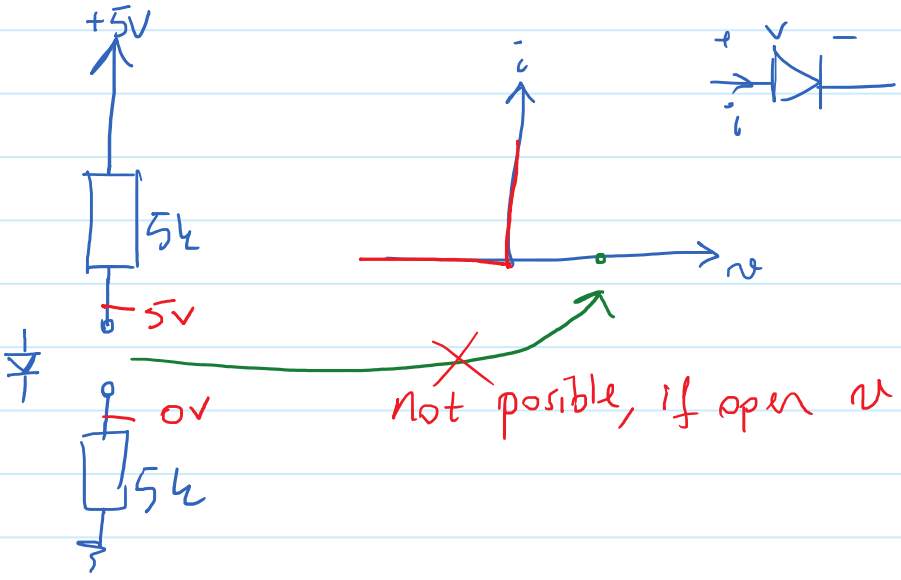
$V_s =$? Write only the answer, no calculations!!!

Q2



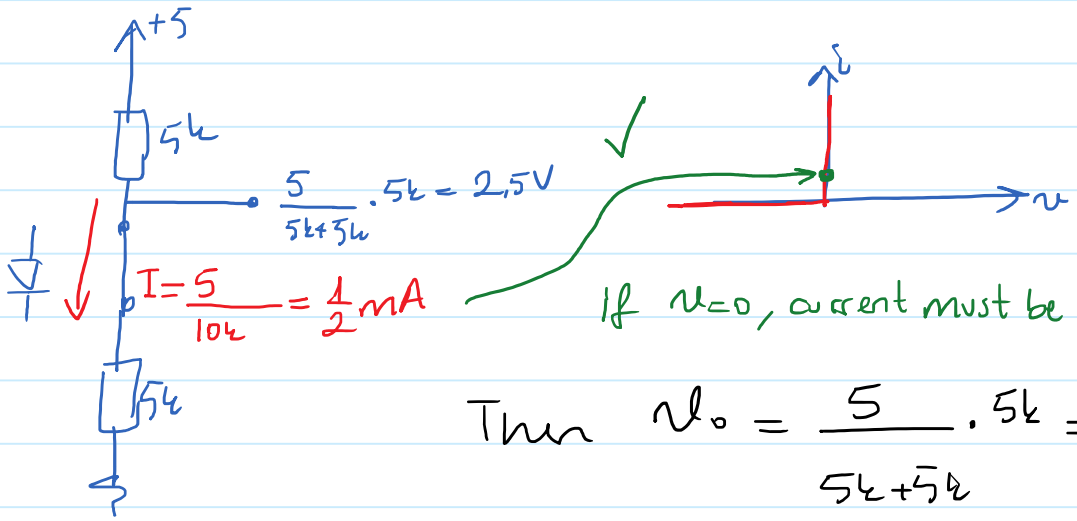
$V_o = ?$

Check if the diode is open circuit



not posible, if open v must be negative

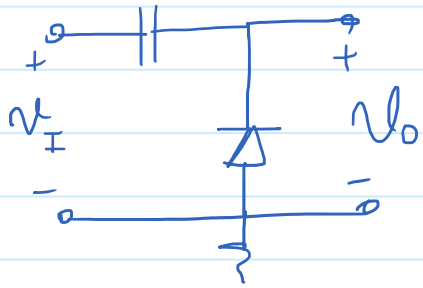
Check if the diode is short circuit.

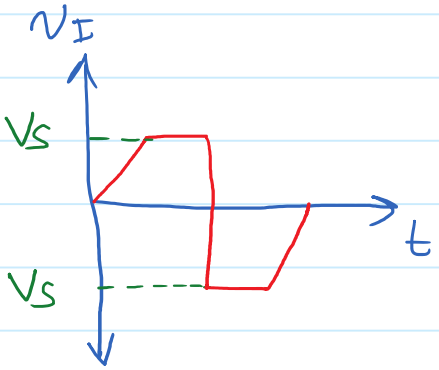


If $v_{=0}$, current must be positive

Thus $V_o = \frac{5}{5k+5k} \cdot 5k = 2.5V$

Q3



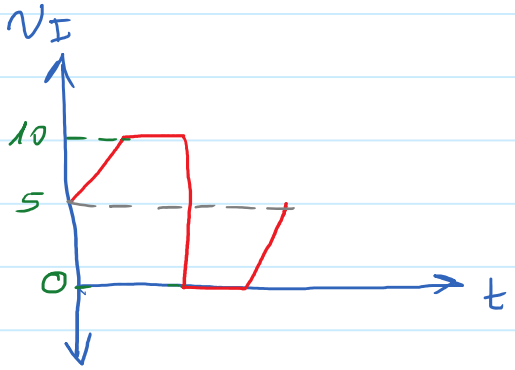


Assuming ided diode characteristic, find the maximum of v_o.

$$\max(v_o) = ?$$

This is a clamping circuit and diode forces v_o to be positive. The minimum of v_o could be zero.

We will shift the v_o up, until its min value is zero.



Then $\max(v_o) = 10V$